



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECR 1213

Network Communication

Task 3 – Choosing the Appropriate LAN Device

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1.0 Introduction to Needed Devices

1. Router

A router is a device that connects computer networks and forwards data packets between them. It acts as a traffic director, determining the best path for data to travel from one network to another. When data is sent through a network, it is divided into smaller packets. These packets are then forwarded from one router to another until they reach their destination. Routers use routing tables or policies to determine the next network for the packet to be sent to. They are connected to multiple data lines from different IP networks, allowing them to direct packets to the appropriate destination. Routers play a crucial role in the functioning of the internet and are essential for communication between networks.

2. Switch

A network switch is a small and important device that centralizes communications among several connected devices such as computers or other network devices in one local area network (LAN) which enables them to talk by exchanging data packets. Network devices are made with the help of switches that carry a large amount of traffic in telecommunications. Switches can be hardware devices that manage physical networks or software-based virtual devices. Besides that, switches have many significant ports, and when data from the network devices arrives at any port, the destination address is examined first, and some checks are also done and then it is processed back to the devices. In the Open Systems Interconnection (OSI) model, a network switch operates on the data-link layer, or Layer 2.

3. Wireless Access Point (WAP)

WAP functions primarily at the data link layer and enables wireless communication using IEEE 802.11 (Wi-Fi) standards. It acts as a central radio transmitter and receiver that bridges wireless client devices to a wired Ethernet network. Modern access points support advanced technologies such as MIMO (Multiple Input Multiple Output), beamforming, MU-MIMO, and WPA3 encryption, which enhance data transmission efficiency, coverage reliability, and network security. In enterprise environments, multiple access points are deployed with centralized management to ensure seamless roaming and load balancing.

4. Modem

Modem (modulator–demodulator) operates at the physical and data link layers of the OSI model and is responsible for converting digital data from network devices into analog or optical signals suitable for transmission over service provider media such as fiber-optic, coaxial cable, DSL, or cellular networks. Conversely, it demodulates incoming signals into digital data that can be processed by routers and end-user devices. The performance of a modem directly impacts bandwidth, latency, and overall network throughput.

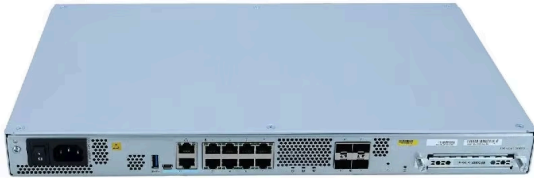
5. Firewall

A firewall is a network security system, available as hardware or software, that monitors and controls incoming and outgoing traffic based on predefined rules. It acts like a security guard, filtering data packets to either accept, reject or drop. A firewall is a network security system, available as hardware or software, that monitors and controls incoming and outgoing traffic based on predefined rules. It acts like a security guard, filtering data packets to prevent unauthorized access, block malicious traffic, protect sensitive information, control network usage and mitigate insider risks. By combining prevention, monitoring, and control, firewalls provide customizable and comprehensive protection against both external and internal threats.

2.0 Table of devices

Devices	Description	Price per unit (RM)	Quantity	Total price (RM)
Router	<u>Cisco ISR 1100 Series (C1111-8P)</u>  Fanless & Quiet: Designed for office/lab environments; runs silently without distracting students during lessons. Enterprise Security: Integrated Cisco Trust Anchor technology protects against counterfeit hardware and ensures secure boot. SD-WAN Ready: Supports Software-Defined WAN, allowing intelligent traffic management between the faculty and the main UTM campus. 8x Gigabit LAN Ports: Provides ample high-speed wired connections for local servers and core switches. IOS XE Software: Runs on the latest open and programmable operating system, making it "4IR ready" for automation.	3,200.00	1	3,200.00
Switch	<u>Cisco Catalyst 9200L-48P-4G</u>  Full PoE+ Capability: Delivers power (740W budget) to Wi-Fi Access Points and IP phones directly through the cable, reducing clutter . Resiliency: Supports Field-Replaceable Units (FRU) for power supplies and fans to ensure continuous operation. Layer 3 Features: Supports routed access, OSPF, and EIGRP stub for efficient network segmentation between labs and admin offices. Enhanced Security: AES-128 MACsec	5,500.00	6	33,000.00

	encryption protects data privacy across the local network. WebUI Management: Easy-to-use interface for monitoring student usage and network health.			
Wireless Access Point	<u>Huawei AP4050DN-HD Access Point</u>   Wi-Fi Standard: 802.11ac Wave 2 (Wi-Fi 5) Max Speed: 2.967 Gbps total (1.275Gbps @5GHz + 1.149Gbps @2.4GHz) MIMO Configuration: 4x4:4 (both 2.4GHz & 5GHz radios) Concurrent Users: 256+ (High-Density optimized) Antennas: Built-in smart antennas (6 dBi gain) Ethernet Ports: 2 x Gigabit Ethernet (1 PoE IN, 1 PoE OUT/LAN) PoE Requirement: 802.3at (PoE+) Dimensions: 220mm x 220mm x 48mm Mounting: Ceiling/wall mount, standard T-bar compatible Features: Smart antenna, beamforming, MU-MIMO, 160MHz channel bandwidth	2,200.00	11	24,200.00
Modem	<u>Huawei HG8240</u> Do not need to buy. The ISP (Unifi, Maxis, or Time) will provide it. Physical Appearance: A small, white or black box, roughly the size of a sandwich. GPON (Gigabit Passive Optical Network): This is the standard technology used by TM/Maxis. It allows	Free (Included with your business fiber subscription)	2	FREE

	one fiber strand to serve multiple buildings via splitters. Speed Capacity (1 Gbps - 2 Gbps): Standard ONUs can handle up to 2Gbps downstream. This is plenty for your office needs. Ports (1x Optical Port (SC/UPC)&4x Gigabit LAN Ports&1-2x Phone Ports (RJ11)): Will only use LAN Port 1 to connect to the FortiGate. The Phone ports are used if you have a fax or analog landline.			
Firewall	<u>Cisco Firepower 1140 NGFW</u>  High Session Capacity: Supports up to 400,000 concurrent sessions, ensuring smooth connectivity for all labs, classrooms, and Wi-Fi users. Powerful Application Control: AVC (Application Visibility & Control) enables real-time monitoring of traffic patterns and identifies over 4,000+ applications for precise security policies. Strong NGFW Performance: Delivers up to 3.3 Gbps Threat Defense throughput, suitable for high-speed lab activities, video conferencing, and cloud-based learning platforms. Advanced Threat Protection: Includes Cisco's Intrusion Prevention System (NGIPS) and Advanced Malware Protection (AMP) for defending against worms, ransomware, and DoS attacks. Scalable for Growth: Designed to handle increasing traffic demands as the Faculty expands by 15% in the coming years. Secure VPN Connectivity: Supports IPsec and SSL VPN, enabling reliable communication between the new building and the main campus network. Flexible Port Options: Comes with 8×	10,000.00	1	10,000.00

	Gigabit RJ-45 ports and 2× SFP fiber ports, ideal for linking access switches and uplinking to the core. Centralized Management: Fully managed through Cisco Firepower Management Center (FMC) or cloud-based Defense Orchestrator for policy automation and real-time analytics. High Availability Support: Allows active/standby failover, ensuring continuous network operation for critical learning environments. Built for Academic Environments: Quiet, compact, and powerful—perfect for secure, high-performance networks in labs and teaching spaces.			
Total				70,400.00

3.0 Reflection

3.1 Are you surprised by the prices? How were you surprised?

We were initially surprised by the cost of the enterprise firewall, which was priced at RM10,000. However, this investment proved to be an optimal choice, balancing our project's budget constraints with the minimum security and performance requirements for our building. The structure is a two-storey, 42m x 26m facility containing four computer labs, one video conference room, and one hybrid classroom. Regarding other hardware costs, the prices for core networking "metal boxes"—such as switches and servers—varied considerably. A notable exception was the Optical Network Terminal (ONT), which was provided at no additional cost as part of our business fiber internet subscription.

3.2 Have you ever considered cost as a factor for choosing networking devices?

Yes, cost was a significant factor in our decision-making process, particularly given our fixed budget of RM 2,000,000. However, rather than simply selecting the cheapest available options (such as consumer-grade routers), we adopted a "Value for Money" approach. We evaluated the cost against the lifespan and reliability of the devices.

For example, while a consumer router costs only RM 500 compared to the Cisco ISR 1100 at RM 3,200, the cheaper option would likely overheat and fail under the load of 1,800 students, leading to network downtime and higher replacement costs in the long run. Therefore, we prioritized Enterprise-grade equipment that, while more expensive upfront, provides the durability, security, and "4IR readiness" required by the Case Study, ensuring we do not need to replace the hardware again in 2-3 years. This strategy ensures we optimize our budget for long-term efficiency rather than short-term savings.

3.3 What are the major differences between the same devices from different brands?

Router:

For the routers used in this project, we have chosen the Cisco ISR 1100 (C1111-8P). The major difference between the Cisco ISR 1100 and other different brands we can compare is the Huawei NetEngine AR610.

Specification	Cisco ISR 1100 (C1111-8P)	Huawei NetEngine AR610
Operating System	Cisco IOS XE (Open & Programmable)	Huawei VRP (Versatile Routing Platform)
LAN Ports	8x Gigabit Ethernet	4x Gigabit Ethernet
Flash Memory	4 GB	1 GB
Fanless Design	Yes (Quiet Operation)	Yes (Quiet Operation)
SD-WAN	Integrated (Viptela/Meraki)	Integrated (Huawei iMaster)

Based on the table above, the Cisco ISR 1100 is a superior router to the Huawei AR610 in terms of hardware capacity and port density. It boasts 4GB of Flash Memory compared to the Huawei's 1GB, allowing it to store larger software images and support more complex routing tables, which is essential for the faculty's long-term growth. Additionally, the Cisco unit offers 8 LAN ports straight out of the box, whereas the Huawei model only offers 4. Most importantly, since our building houses a Cisco Networking Lab, using a Cisco router for the building's main gateway provides an excellent real-world learning opportunity for students to study the actual equipment running their network

Switch:

For the switches used in this project, we have chosen the Cisco Catalyst 9200L-48P. The major difference between the Cisco Catalyst 9200L and other different brands we can compare is the Huawei S5735-L48P4X.

Specification	Cisco Catalyst 9200L	Huawei S5735-L48P4X
Stacking Tech	StackWise-80 (Hardware Stacking)	iStack (Intelligent Stacking)
PoE Power Budget	740W (Full PoE+)	380W (Standard)
Uplinks	4x 1G Fixed Uplinks	4x 10G Fixed Uplinks
Management	Cisco DNA Center	Huawei iMaster NCE

Reliability	High (Field Replaceable Fans/Power)	Moderate (Fixed Fans/Power)
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Based on the comparison table above, the Cisco Catalyst 9200L is selected primarily for its reliability and power capabilities. The Cisco model offers a robust PoE budget of 740W, which is significantly higher than the Huawei's standard 380W. This high power budget is critical for supporting our high-performance Wi-Fi 6 Access Points and security cameras without risking power outages. Furthermore, the Cisco switch features Field Replaceable Units (FRU) for fans and power supplies, meaning if a fan breaks, we can replace just the fan without replacing the whole switch. While the Huawei switch offers faster 10G uplinks, the Cisco Catalyst 9200L's compatibility with the "4IR" industry standard (Cisco DNA) and its superior power handling make it the most suitable selection for our faculty building.

Wireless Access Point (WAP):

For the WAP used in this project, we have chosen the Cisco Aironet 2802i Series Access Point Wireless AIR-AP2802I-K-K9.

Specification	Cisco Aironet 2802i (AIR-AP2802I-K-K9)	Huawei AP4050DN-HD
Wi-Fi Generation	802.11ac Wave 2 (Wi-Fi 5)	802.11ac Wave 2 (Wi-Fi 5)
Max Theoretical Speed	2.9 Gbps (5GHz, 3x3:3 MIMO)	2.967 Gbps (5GHz, 4x4:4 MIMO)
Dual-Radio Design	Yes (2.4GHz & 5GHz)	Yes (2.4GHz & 5GHz)
Ethernet Uplink	1 x 1 Gbps RJ45 (PoE+)	2 x 1 Gbps RJ45 (1 for PoE+ in, 1 for out/passthrough)
Smart Features	Cisco Hyperlocation	Huawei Smart Antenna (beamforming, coverage optimization)
Management	Catalyst 9800 / AireOS WLC	Agile Controller-based (AC6005, etc.) or Cloud-based (iMaster NCE)
Typical Street Price	RM 2,310.00	RM 2,200.00

From the table above we can observe that both WAP have the same technology lifecycle. Both are mature Wi-Fi 5 products. Their R&D costs are fully amortized, so pricing is competitive to clear inventory and win market share. To displace entrenched competitors like

Cisco, Huawei often offers more features (4x4 MIMO, dual ports) at the same price or similar features at a significant discount. In addition, Huawei's wireless controller licensing and support costs are frequently lower than Cisco's DNA licensing model, making the overall project cheaper. According to the comparison table above we can highlight:

- 1) Equivalent 802.11ac Wave 2 performance.
- 2) Technical advantages (4x4 MIMO, dual Ethernet ports).
- 3) A lower total project cost (AP unit price + controller licensing).

To conclude, the Huawei AP4050DN Series (AP4050DN-E) is the price-competitive, generation-equivalent product that is more worth buying compared to Cisco Aironet 2802i (AIR-AP2802I-K-K9).

Modem (ONT):

For the modem we chose Huawei HG8240 optical network terminal, the difference between Huawei and ZTE is negligible. In "Bridge Mode," they both act as a simple "dumb pipe" to convert light to electricity. Hence, we are going to compare it with the ZTE F600.

Specification	Huawei (HG8240)	ZTE (F600)
Heat Management	Efficient passive cooling design with optimized vent placement; generally maintains lower operating temperatures.	Tends to run hotter under sustained load; passive cooling is less effective compared to Huawei.
Stability and Uptime	Very high; known for extended operation without freezing, often months without requiring a reboot.	Moderate; older models may require periodic reboots (every 3–6 months) due to overheating-related issues.
Bridge Mode Reliability	Highly reliable; operates as a transparent layer 2 bridge with consistent performance and no packet interference.	Generally functional, but occasional issues with LAN Port 1 link negotiation (speed/duplex) may require manual intervention.

Huawei HG8240 has better overall stability and reliability in a Bridge Mode deployment. While both modems function adequately as "dumb pipes," the Huawei model excels with better passive cooling, a longer uptime without needing reboots, and more consistent, trouble-free performance in passing traffic directly to a router (like a FortiGate). Therefore, the Huawei HG8240 is the superior and recommended choice over the ZTE F600.

Firewall:

For the firewall used in this project, we have chosen the Cisco Firepower 1140 NGFW. The major difference between the Cisco Firepower 1140 and other different brands we can compare is the Fortinet FortiGate 60F.

Specification	Cisco Firepower 1140	Fortinet FortiGate 60F
Firewall Throughput (Gbps)	3.3	10
IPS/Threat Prevention (Gbps)	3.5	1.4
VPN Throughput (Gbps)	1.2	6.5
Concurrent Sessions	400K	-
Form Factor/Ports	1RU, 8x RJ45 + 4x SFP	Desktop, 10x GE

Based on the comparison table above, the Cisco Firepower 1140 is selected primarily for its strong balance of performance and security features tailored for our project. It delivers higher IPS throughput at 3.5 Gbps and supports up to 400,000 concurrent sessions, which are essential for handling the high volume of connections and complex traffic typical in academic environments with many users and devices. The device offers extensive URL filtering with over 80 categories and robust malware analytics, providing comprehensive protection for diverse computing needs. While the Fortinet FortiGate 60F excels in raw firewall throughput, the Cisco Firepower 1140's superior session capacity and advanced threat prevention capabilities make it a better fit for the demanding and scalable security requirements of the building.

4.0 Meeting minutes

4.1 Meeting minutes #1

DATE/TIME		4 December 2025 19:00	
LOCATION		Online(Google Meet)	
AGENDA		1. Analysis Task 3 requirements 2. Identify all device types required 3. Research suitable brands and models 4. Task distribution	
Meeting MC		Tey Xin Yi	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Tey Xin Yi		19:00	
Lee Hui Zhen		19:00	
Tan Hui Shan		19:00	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Understanding Task 3 Requirements	- Tey explained that Task 3 requires listing all LAN devices, researching prices, and comparing brands. - All members reviewed device categories (router, switch, WAP, modem and firewall)	Tey (4/12)
2	Identifying Devices Needed	- Lee suggested listing enterprise-grade devices instead of home-grade. - Tan added labs require high-performance equipment due to 30 PCs per lab.	Lee & Tan (4/12)
3	Researching Device Brands	- Lee proposed Cisco for consistency with the Cisco Networking Lab. - Tan suggested comparing Huawei as a secondary option. - Tey suggested comparing Asus also.	All members (4/12)

4	Task Distribution	- Task distribution assigned as below: 1. Lee-research for router and switch 2. Tey-research for wireless access point and modem 3. Tan-research for firewall and prepare meeting minutes	All members (4/12)
5	Meeting ended	22:00	

4.2 Meeting minutes #2

DATE/TIME		6 December 2025 20:00	
LOCATION		Online(Google Meet)	
AGENDA		1. Review shortlisted devices 2. Comparing between different brands 3. Select final devices and quantities	
Meeting MC		Lee Hui Zhen	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Tey Xin Yi		20:00	
Lee Hui Zhen		20:00	
Tan Hui Shan		20:00	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Router Selection	- Lee recommended Cisco ISR 1100 due to fanless design, SD-WAN support, and 8 Gigabit LAN ports. - Team members agreed Cisco aligns with the networking lab environment.	Lee (6/12)
2	Switch Selection	- Lee compared Cisco Catalyst 9200L and Huawei S5735-L. - Cisco chosen for higher PoE budget (740W), FRU support, and enterprise stability.	Lee (6/12)
3	WAP Selection	- Tey recommended Huawei AP4050DN-HD for technical advantages (4x4 MIMO, dual Ethernet ports) and lower cost compared to other modem with similar features. - Tan confirmed 11 APs required based on coverage.	Tey (6/12)
4	Modem Selection	Tey confirm modem (ONT) provided free by ISP; no purchase required.	Tey (6/12)
5	Firewall	Tan proposed Cisco Firepower 1140 due to 400K concurrent sessions, 3.3 Gbps	Tan (6/12)

	Selection	throughput, and AMP/IPS. Team members agreed 1 unit is enough.	
6	Quantity Confirmation	- Router: 1 - Switch: 6 - WAP: 11 - Modem: 2 - Firewall: 1	All members (6/12)
7	Meeting Ended	22:00	

5.0 References

- Wikipedia contributors. (2025, November 30). Router (computing). Wikipedia.
https://en.wikipedia.org/wiki/Router_%28computing%29
- GeeksforGeeks. (2025, October 6). Router in computer networks. GeeksforGeeks.
<https://www.geeksforgeeks.org/computer-networks/introduction-of-a-router/>
- Wikipedia contributors. (2025b, December 3). Network switch. Wikipedia.
https://en.wikipedia.org/wiki/Network_switch
- Yasar, K., & Burke, J. (2023, March 7). network switch. Search Networking.
<https://www.techtarget.com/searchnetworking/definition/switch>
- Cisco ISR1100 and ISR1100X Series Routers Data Sheet. (2024, October 17). Cisco.
<https://www.cisco.com/c/en/us/products/collateral/routers/1000-series-integrated-services-routers-isr/datasheet-c78-742893.html>
- C1111-8P. (2020). Router-Switch.com. <https://www.router-switch.com/c1111-8p.html>
- Huawei NetEngine AR610 Series Enterprise Routers Datasheet. (n.d.). Huawei Enterprise.
<https://e.huawei.com/en/documents/products/enterprise-network/e1fc81f9c69d43eb96d1190f77711067>
- Cisco Catalyst 9200 Series Switches Data Sheet. (2025, October 21). Cisco.
<https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-9200-series-switches/nb-06-cat9200-ser-data-sheet-cte-en.html>
- C9200L-48P-4G-E Price - Cisco Switch Catalyst 9200. (2020). Router-Switch.com.
<https://www.router-switch.com/c9200l-48p-4g-e.html>
- Huawei CloudEngine S5735-L Series Switches Datasheet. (n.d.). Huawei Enterprise.
<https://e.huawei.com/en/documents/products/enterprise-network/a7d76a8a16614cefb9943336bb224d69>
- GeeksforGeeks. (2025, November 20). Introduction of Firewall. GeeksforGeeks.
<https://www.geeksforgeeks.org/computer-networks/introduction-of-firewall-in-computer-network/>
- Cisco Firepower 1000 Series Data Sheet. (2024, October 17). Cisco.
<https://www.cisco.com/c/en/us/products/collateral/security/firepower-1000-series/datasheet-c78-742469.html>

Router-Switch. (n.d.). Cisco Firepower 1140 NGFW Appliance Datasheet. Router-Switch.
<https://www.router-switch.com/pdf2html/pdf/fpr1140-ngfw-k9-datasheet.pdf>

ManualMachine. (n.d.). Cisco Firepower 1140 User Manual. ManualMachine.
<https://manualmachine.com/cisco/firepower1140/7677427-user-manual/>

Huawei Technologies Co., Ltd. (n.d.). EchoLife HG8240. (2024, April 8). Huawei Enterprise Support.
<https://support.huawei.com/enterprise/en/access-network/echolife-hg8240-pid-4977718>

ZTE Corporation. (n.d.). ZXHN F660.(2024, April 8) ZTE Product Index.
https://www.zte.com.cn/global/product_index/smart_home_en/ont/zxhn-f660/zxhn-f660.html

Huawei Technologies Co., Ltd. (n.d.). *AP4050DN-HD*. (2024, April 8). Huawei Enterprise Support. <https://support.huawei.com/enterprise/en/wlan/ap4050dn-hd-pid-21968848>